

Model interpretability write-up

Task – 4 : Week 2

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1. Data Science models used for NHS clinical audience. :

The model used for this analysis is a **Random Forest**. In clinical terms, this is like having a hundred different "Diagnostic Trees." Each tree asks a series of "Yes/No" questions about a patient's data to reach a conclusion.

By combining the results of these hundred different pathways, the model provides a highly stable prediction.

2. Key findings :

By analyzing the anxiety_depression_data.csv, the model identified the top three "indicators" that most heavily influence a patient's anxiety levels:

- **Sleep Quality (Top Indicator):** The number of sleep hours was the most significant predictor. The model suggests that even small changes in sleep patterns can drastically shift a patient's mental health risk profile.
- **Physical Activity:** Patients with low physical activity hours were consistently flagged as higher risk. This validates the clinical importance of lifestyle interventions.
- **Age and Life Stage:** The model found that age acts as a "weighting factor," meaning the impact of stress or lack of sleep varies significantly depending on how old the patient is.

3. Interpretability :

A common concern with computer models is that they are "Black Boxes"—meaning we see the answer but don't know *why* the computer chose it. For an NHS setting, we prioritize Interpretability:

- **Transparency:** We can see which specific features (like Financial Stress or Loneliness) pushed the risk score up.
- **Feature Importance:** As seen in our analysis, the model doesn't just say "this patient is anxious"; it tells us that Age and Sleep were the primary reasons for that specific prediction.

4. Clinical Application :

For a GP or mental health practitioner, this model acts as a Decision Support Tool:

1. **Early Warning:** Identifying patients who might not "**look**" **high-risk** in a quick 10-minute consultation but whose underlying data (high work stress + low sleep) matches the profile of high-anxiety cases.
2. **Personalized Care:** If the model shows that "Financial Stress" is the main driver for a specific group, the clinical response might shift from medication toward social prescribing or financial counseling.
3. **Resource Allocation:** Directing high-intensity therapy to those the model identifies as having the most "unstable" indicators (e.g., poor sleep combined with high loneliness).

5. Summary Table for Ward Use :

Indicator	Model Insight	Clinical Action
Sleep Hours	High sensitivity; primary driver of anxiety.	Prioritize sleep hygiene in care plans.
Loneliness Score	Significant "hidden" driver of depression.	Consider social prescribing/community groups.
Physical Activity	Strong protective factor.	Encourage mobility and exercise where possible.

6. Conclusion :

The analysis of the **1,200** patient records in our dataset demonstrates that data science models can do more than just "predict"—they can provide clear, actionable insights into the underlying drivers of patient mental health.

By moving away from a "one-size-fits-all" approach, this model highlights that for the current patient cohort, Sleep Hours, Age, and Physical Activity are the three most critical pillars of anxiety and depression risk.

Key Takeaways for the Clinical Team:

- Predictive Insight vs. Traditional Observation: While clinical intuition is invaluable, the model identifies "high-risk" combinations of factors (such as the compounding effect of low sleep and high loneliness) that may not be immediately obvious during a standard consultation.
- Targeted Interventions: The model's "Feature Importance" rankings allow the Trust to prioritize interventions that have the highest mathematical likelihood of improving patient outcomes—specifically sleep hygiene programs and social prescribing for loneliness.